



Ben-Gurion University of the Negev
The Faculty of Natural Sciences
The Department of Computer Science

Expected-Return Inference over Complete Policies

Thesis submitted in partial fulfillment of the requirements
for the Master of Sciences degree

Eden Havri

Under the supervision of **Ronen Brafman**

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Abstract

We study generalized Bayesian inference over complete deterministic policies using the expected return of a stochastic finite-horizon Markov decision process. The Gibbs target is defined before choosing an approximation: environment randomness is averaged before the value is exponentiated. This established policy-search formulation is distinct from weighting individual trajectories by exponentiated realized return.

The current practical method learns a factorized distribution over complete policy tables. A separately costed PPO guide initializes the distribution; direct variational refinement then uses simulator returns, a leave-one-out baseline, and an exact policy-space entropy term. We justify its visited-state-masked score gradient and verify the actual implementation against exact differentiation on enumerable instances. One rollout per sampled policy is

sufficient for gradient correctness, although allocation between policy draws and repeated rollouts affects finite-budget learning. The method samples fresh policies between updates and commits to one table during each evaluation episode. It has no particle-resampling stage.

The empirical design separates gradient correctness, variational family error, and control performance. Strict interaction caps include guide training and refinement; native stochastic execution and committed-policy execution are both reported. Preliminary GridWorld studies favor one rollout per policy in a medium-map allocation test. A failed hard run exposes insufficient exploration; reallocating the same budget toward guide training improves its subsequent control without eliminating the gap between execution rules. A subsequent sixteen-run BGU calibration retains all independent-guide outcomes, including a medium-map failure in one of three primary-map seeds. Assisted PPO has higher return in seven of eight map-seed pairs. A separately bounded medium-map reallocation recovers the selected failed seed and avoids collapse in two fresh seeds; assisted PPO still has higher native return on both fresh seeds. This limited follow-up does not establish general robustness. These are development results, not confirmatory superiority or general posterior-fidelity claims.

The thesis also documents alternative policy samplers and their failed development configurations. An exact small-MDP diagnostic separates family error from control quality. The full five-baseline comparison and held-out evaluation remain necessary before final conclusions.

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Chapter 1

Introduction

Planning in a stochastic environment requires two conceptually separate operations. First, a policy is evaluated by averaging its return over the environment's transition randomness. Second, a decision rule selects among policies. Conflating these operations changes the objective: exponentiating a single realized return before averaging is generally not equivalent to exponentiating the expected return.

This manuscript develops a policy-inference formulation in which the target is defined before an approximation algorithm is chosen. The target assigns mass to complete deterministic policies according to expected return. Its modes are therefore expected-return maximizers, while a finite inverse temperature retains a distribution over alternatives. The construction is a generalized Gibbs posterior over decisions, not a posterior belief about unknown transition parameters. Related free-utility formulations also trade expected utility against information cost [1], and recent general Bayesian policy learning

updates distributions over decision rules from a loss [2]. Our setting differs by targeting finite-horizon MDP policies and by making the expected-return order of operations explicit.

Control-as-inference methods provide a broad connection between optimal control and probabilistic inference [3]. In stochastic dynamics, however, the choice of graphical model and optimality likelihood determines whether the resulting objective is risk-neutral, risk-sensitive, or maximum-entropy. We therefore do not treat “inference” as sufficient to identify the control objective. Expected-return Gibbs policy search and Metropolis–Hastings inference over policies have direct precedent in Wingate et al. [4], who also discuss replica exchange. Neither the exponential target, its mode preservation, nor policy-space MCMC is claimed here as a new inference principle. The research question is how to approximate this target at a defensible computational cost, and what control behavior follows from committing to a sampled policy. The current practical candidate uses direct variational learning; corrected MCMC and fixed-bank SMC provide separately assessed alternatives. Preliminary development comparisons motivate, but do not replace, held-out validation.

The study has five components:

1. It specifies an expected-return Gibbs measure over complete policies, including its reference measure, temperature, and execution semantics.
2. It gives exact finite-policy oracles and variance-sensitive diagnostics that detect substitution of an exponentiated-trajectory objective.
3. It implements direct policy-space variational learning with an unbiased masked score-function gradient, independent baseline, and exact policy-

table entropy term.

4. It separates variational family error, fixed-tape approximation, and MCMC mixing, retaining corrected guided MCMC and replicated SMC as diagnostic alternatives rather than conflating their guarantees.
5. It defines a fail-fast comparison against five published algorithmic families with both policy-commitment and native execution semantics, strict total interaction caps, and separately labeled exploration aids.

Exact small-instance checks validate target definitions, gradient expectations, and sampling kernels before scaling. They do not establish full posterior agreement for the variational family. The protocol is specified before confirmatory results; this draft labels development findings as preliminary.

Chapter 2

Problem Setting and Notation

2.1 Finite-horizon stochastic control

Let an episodic Markov decision process be $\mathcal{M} = (\mathcal{S}, \mathcal{A}, T, R, \rho_0, H, \gamma)$, where $T(\cdot \mid s, a)$ is a transition kernel, $R(s, a, s')$ is a bounded reward, ρ_0 is the initial-state distribution, and H is the rollout horizon. The discount is $\gamma \in (0, 1]$; the current GridWorld study uses $\gamma = 1$.

For a deterministic policy $\pi \in \Pi$, a trajectory $\tau = (s_1, a_1, \dots, s_{H+1})$ is generated by

$$s_1 \sim \rho_0, \quad a_t = \pi(s_t), \quad s_{t+1} \sim T(\cdot \mid s_t, a_t).$$

Its truncated return and the policy value are

$$G_H(\tau) = \sum_{t=1}^H \gamma^{t-1} R(s_t, a_t, s_{t+1}), \quad J_H(\pi) = \mathbb{E}_{\tau \sim p(\cdot \mid \pi)}[G_H(\tau)]. \quad (2.1)$$

The implemented policy class is stationary in the physical state. Time-to-go is deliberately not appended to the observation, so recurrent states at different remaining horizons can be aliased. Every learned and inferred method shares this restriction. Finite-horizon dynamic programming is reported as a nonstationary upper control, not as a like-for-like competitor. A future time-indexed study must augment the state for every method rather than only for the proposed approach.

2.2 Policy-level randomness

A distribution over deterministic policies is different from a Markov stochastic policy. Sampling one deterministic policy at the start of an episode creates correlations across states and revisits. Sampling a new action from a statewise marginal on every visit removes those correlations. Both execution rules can be studied, but they must not be treated as interchangeable. Our primary rule is *sample then commit*: draw all state actions defining one deterministic policy before evaluation and preserve them throughout the episode. Per-visit marginal execution and the statewise maximum-probability policy are secondary diagnostics.

Assumption 2.2.1 (Reference measure). *The deterministic policy class has a specified probability measure μ . For a finite tabular class, μ may be uniform. For a structured or continuous class, the choice of μ is part of the model and must be reported.*

Chapter 3

Expected-Return Policy Target

For inverse temperature $\beta > 0$, define the target measure

$$P_\beta(d\pi) = \frac{\exp(\beta J_H(\pi))}{Z_\beta} \mu(d\pi), \quad Z_\beta = \int_{\Pi} \exp(\beta J_H(\pi)) \mu(d\pi). \quad (3.1)$$

Bounded rewards and a finite reference measure make Z_β finite. The parameter β carries the inverse units of reward, so a change of reward units must be accompanied by the inverse change in β . The pair (μ, β) is part of the problem specification: neither is identified by simulator data alone. For the tabular experiments, μ is uniform over complete stationary deterministic policies.

Proposition 3.0.1 (Mode preservation). *Every mode of P_β with respect to a uniform finite reference measure maximizes J_H , and every expected-return maximizer is a mode.*

Proof. The exponential function is strictly increasing, and the normalizing constant does not depend on π . □

The order of expectation and exponentiation is essential. In general,

$$\exp(\beta \mathbb{E}[G_H]) \neq \mathbb{E}[\exp(\beta G_H)]. \quad (3.2)$$

The right-hand side defines an entropic-risk quantity and places additional weight on return variability. By Jensen’s inequality it is at least the left-hand side for $\beta > 0$, with strict inequality for nonconstant return. It is not an estimator identity for the target in Equation (3.1).

The target also has the variational characterization

$$\log Z_\beta = \sup_Q \{ \beta \mathbb{E}_{\pi \sim Q}[J_H(\pi)] - \text{KL}(Q \parallel \mu) \}. \quad (3.3)$$

Thus the regularization acts on a distribution over complete policies. It is not the same as adding independent action entropy at every visited state.

Equation (3.1) may be read as a generalized posterior whose loss is $-J_H$. This interpretation is useful algorithmically, but it does not imply epistemic uncertainty calibration or Bayesian learning of the MDP. At $\beta \rightarrow \infty$, mass concentrates on expected-return maximizers; at finite β , output quality and diversity depend on both the temperature and the policy reference measure.

Chapter 4

Exact Validation Framework

Correct implementation of a gradient or MCMC transition does not establish that the intended density was specified. Validation therefore starts with finite MDPs for which all $|\mathcal{A}|^D$ stationary deterministic policies can be enumerated, where D is the number of decision states. Finite-horizon dynamic programming computes $J_H(\pi)$ for every policy, and log-sum-exp normalization gives the exact target probabilities, partition function, and action marginals in Equation (3.1).

4.1 Target-substitution diagnostic

The smallest diagnostic contains two policies with equal expected return but different return variance. Under a uniform reference measure, Equation (3.1) assigns them equal probability. A method that averages $\exp(\beta G_H)$ instead assigns more mass to the variable policy. This is a deterministic pass-or-fail

test of the order of operations rather than a performance benchmark.

Additional exact tests cover state revisitation, absorbing states, finite horizons, policy memoization, common random numbers, Metropolis–Hastings proposal ratios, replica swaps, particle ancestry, and score-function gradients. In particular, empirical frequencies of both single-chain and replica-exchange samplers must agree with the enumerated target on tiny MDPs.

4.2 Fail-fast gates

Experiments advance through checks appropriate to their inference engine. Mechanics, target definition and cost apply to every implementation; finite-bank and mixing checks apply only to the corresponding sampling variants. For variational inference, gradient correctness, optimization progress and family error replace a requirement to pass MCMC transport diagnostics:

- G1. **Mechanics.** Transition rows, rewards, termination, horizons, seeded replay, artifact writing, and budget counters pass unit tests.
- G2. **Target.** Enumeration verifies exact probabilities, modes, marginals, local and whole-policy acceptance ratios, and swap acceptance ratios.
- G3. **Approximation.** For variational inference, compare exact and sampled gradients, measure optimization progress, and characterize the restricted-family gap on enumerable instances. For fixed-bank methods, increasing the immutable tape-bank size reduces sample-average target error on repeated small problems; a failure stops simulator-only scaling.
- G4. **Mixing (MCMC only).** A one-chain rescue check first requires an

accepted, policy-changing whole-policy refresh in the upper quarter of the inverse-temperature ladder, an accepted final-edge swap, more than one cold-reaching walker, and at least two sampled cold-replica policies. A cheap two-chain ladder-flow probe then requires post-burn acceptance of at least 0.20 on every edge and an endpoint transition in each chain. Passing this probe authorizes, but does not replace, the four-chain mixing run. That full gate requires split return $\hat{R} \leq 1.05$, per-chain return ESS of at least 100, flow through every ladder edge and in both directions, between-chain coordinate agreement, and within-chain policy movement. High local acceptance or either passing fail-fast probe alone is not evidence of mixing.

G5. **Cost.** Three short runs establish memory, wall-clock time, and value-evaluation or simulator-step rates before a large job array is submitted.

The final thresholds for effective sample size and target error are frozen after pilot calibration and before confirmatory seeds are inspected. Failed, timed-out, and numerically invalid runs remain in the accounting.

Chapter 5

Expected-Return Policy Inference

5.1 Practical candidate: a distribution over complete policies

The current practical candidate is direct variational policy inference. It is distinct from the corrected MCMC and particle alternatives retained later in this chapter. The target is chosen first; an inference engine must then state its approximation and information requirements. MCMC mixing is not a prerequisite for evaluating a variational optimizer, just as a correct variational gradient is not evidence of posterior equality.

Let $\mathcal{D}$ be the decision-state set and let

$$q_\theta(\pi) = \prod_{s \in \mathcal{D}} q_{\theta,s}(\pi(s)) \quad (5.1)$$

be a categorical mean-field family. Drawing a table from this family makes its state actions independent at draw time; retaining that table makes repeated visits to the same state use the same action. Independence between coordinates therefore does not imply action resampling during execution.

The learning objective is

$$\mathcal{L}(\theta) = \beta \mathbb{E}_{\pi \sim q_\theta} [J(\pi)] - \text{KL}(q_\theta \| \mu), \quad (5.2)$$

$$\log Z_\beta - \mathcal{L}(\theta) = \text{KL}(q_\theta \| P_\beta). \quad (5.3)$$

Equation (5.3) follows by substituting the log target density and taking its expectation under q_θ . Maximizing this ELBO minimizes reverse KL within the chosen family. It does not remove the family restriction, ensure a global optimum, or make the learned distribution a calibrated posterior about unknown environment parameters.

For the uniform reference used here, the regularizer is tractable:

$$-\text{KL}(q_\theta \| \mu) = - \sum_{s \in \mathcal{D}} \sum_a q_{\theta,s}(a) \log q_{\theta,s}(a) - |\mathcal{D}| \log |\mathcal{A}|. \quad (5.4)$$

It concerns every table coordinate, including those not visited by a particular rollout. It is not the visitation-weighted entropy bonus commonly used to encourage action-level exploration.

5.2 The actual gradient estimator

Draw independent complete policies $\pi_1, \dots, \pi_N$. For each, draw K independent simulator rollouts and compute

$$\bar{G}_i = \frac{1}{K} \sum_{k=1}^K G_H(\pi_i, \xi_{ik}), \quad b_i = \frac{\beta}{N-1} \sum_{j \neq i} \bar{G}_j \quad (N > 1).$$

For a single configured policy sample the baseline is zero. Let M_{is} indicate that $\pi_i(s)$ is used before the horizon in at least one of the K rollouts. Merely arriving at s at the final time does not reveal its selected action and does not include that coordinate in the mask. The implementation uses

$$\hat{g} = \frac{1}{N} \sum_i (\beta \bar{G}_i - b_i) \sum_{s \in \mathcal{D}} M_{is} \nabla_\theta \log q_{\theta,s}(\pi_i(s)) - \nabla_\theta \text{KL}(q_\theta \parallel \mu). \quad (5.5)$$

The return and leave-one-out baseline are held constant when differentiating the score-function surrogate [5]. The entropy/KL contribution is differentiated exactly. Each selected policy action contributes at most one score term, irrespective of how often that state is revisited.

Proposition 5.2.1 (Unbiased masked policy gradient). *For the independent mean-field policy sampler, parameter-independent MDP dynamics, independent simulator rollouts, and the baseline above, $\mathbb{E}[\hat{g}] = \nabla_\theta \mathcal{L}(\theta)$.*

Proof. The finite policy space and bounded returns permit differentiation under the sum and rollout expectation. Without masking, the score-function identity gives the expected-return gradient. Fix a state s , condition on all other policy coordinates and simulator random numbers, and compare possible

values of $\pi(s)$. Before the first visit to s , that action has not been used, so the event of visiting s is independent of its sampled action. On the complementary event, every rollout return is independent of $\pi(s)$. Since the categorical score has zero expectation, deleting the unvisited return-score term leaves the expected gradient unchanged. The same argument gives zero mean for the masked score itself. Each b_i depends only on other independent policies and rollouts, so its product with that score has zero expectation. Finally, the exactly differentiated KL term is the required regularizer gradient. $\square$

The union mask for $K > 1$ preserves the argument: if the state is absent from all rollouts, none of their returns depends on its action. A rollout that does not visit the state may add variance when another rollout does visit it, but does not invalidate the expectation identity. The proof does not automatically extend to dependent policy draws, nonfactorized policy families, or dynamics parameterized by the same θ .

Proposition 5.2.2 (Variance relative to the full-table score). *At fixed θ , under the assumptions above, let $\hat{g}_{\text{full}}$ be Equation (5.5) with every M_{is} replaced by one, using the same tables, returns, baseline and exact KL derivative. Then*

$$\text{Cov}(\hat{g}) \preceq \text{Cov}(\hat{g}_{\text{full}}). \quad (5.6)$$

Proof. Let $\mathcal{F}$ contain all observed trajectories, their returns and the sampled table actions actually used in those trajectories. It excludes unrevealed table entries. Factorization and parameter-independent simulation leave each unused action distributed according to its original categorical law conditional

on $\mathcal{F}$; its conditional mean score is zero. The used-action scores and whole-return leave-one-out coefficients are $\mathcal{F}$ -measurable. The KL derivative is deterministic at fixed θ . Hence

$$\hat{g} = \mathbb{E}[\hat{g}_{\text{full}} \mid \mathcal{F}].$$

For finite second moments, the law of total covariance gives

$$\text{Cov}(\hat{g}_{\text{full}}) - \text{Cov}(\hat{g}) = \mathbb{E}[\text{Cov}(\hat{g}_{\text{full}} \mid \mathcal{F})] \succeq 0.$$

Using the union across the K rollouts is essential: an action revealed by one rollout cannot be averaged out merely because another did not visit it. $\square$

This is the standard conditional-expectation variance argument applied to the policy-table score, not a new inference principle. It guarantees no greater variance in any linear projection than the otherwise identical full-table estimator. It does not compare different baselines, rollout allocations, return-to-go estimators, optimizer trajectories or final control quality. Four exact-enumeration checks of the production estimator, covering $K = 1, 2$ and the baseline on or off, verify both the common mean and the positive-semidefinite covariance difference on the diagnostic MDP.

One rollout per policy is therefore sufficient for correctness. The reward signal is linear in G_H , so replacing J by an unbiased rollout preserves the gradient in expectation. This is different from replacing $e^{\beta J}$ by an exponentiated sampled return. Choosing K is a cost–variance trade-off: more rollouts per policy leave fewer independent policy draws or updates under a common interaction

budget. The development comparison selects $K = 1$ for subsequent validation; it is not a universal optimum claim.

Exact tests enumerate two policy draws and all stochastic visitation outcomes on a small MDP. With $K = 1, 2$ and the baseline enabled or disabled, the expected production gradient agrees with exact ELBO differentiation to 10^{-12} . These validate the estimator, not convergence on large maps.

5.3 Initialization, training and deployment

PPO-Clip first trains a statewise guide under a separately recorded allowance. Its positive clipped probabilities initialize the variational logits. The guide is an initialization, not a replacement reference measure. Current assisted development runs add $1/\sqrt{n(s')}$ to each guide-training transition, where $n(s')$ is the incremented count of arrivals at its next state, including terminal arrivals. The coefficient is 1.0. Refinement and evaluation use only original environment rewards. A comparison that supplies this aid only to the proposed method cannot attribute its entire performance gap to inference.

Each refinement iteration draws fresh tables and independent rollout noise, forms Equation (5.5), and updates logits with Adam. No genealogy, re-sampling weight, or sampled policy persists between updates; the learned distribution parameters and optimizer state do persist. Historical development used $N = 32$. The frozen confirmatory configuration uses $N = 8$, $K = 1$, $\beta = 64$, and learning rate 0.01, with a guide fraction of 0.5 under the common 200k-transition cap. Both variants use the same temperature and policy batch

size. Development configuration selection and fresh-map validation precede this choice; no protected outcome is used to choose it. Guide allocation is calibrated separately from the expected-return target.

At deployment, one table is sampled and retained for an episode. The native action-resampling marginal and deterministic marginal mode are evaluated as distinct controls. Commitment can change behavior without improving it; a single sampled wrong turn can persist on every revisit. Chapter 6 reports a large hard-map gap between these execution rules.

5.4 A strict total simulator allowance

The production runner includes actual PPO-guide usage in the total cap C ; refinement receives the remainder. Before drawing a minibatch it reserves the worst-case NKH transitions. It reduces N to fit the remaining allowance or stops when fewer than two policies fit, preserving the leave-one-out baseline. A configuration explicitly requesting $N = 1$ may use one sample with a zero baseline. No long sampled trajectory is selectively discarded or truncated to satisfy the cap.

The unchanged complete-episode REINFORCE, PPO, SAC and CEM trainers can overshoot their stopping threshold by at most $H - 1$ transitions. Supplying threshold $C - H + 1$ therefore enforces the cap without changing their update equations. Double Q-learning already stops per transition. Actual usage and unused allowance are reported; a common maximum is not an assertion of identical consumption or equal wall time. Exact offline evaluation is costed

separately and never feeds into training. Historical MCMC/SMC branches retain their older guide-only allowance and reject strict mode rather than silently claiming an unimplemented total cap.

5.5 Family error without poor control

Consider a two-step MDP. At the start, action 0 terminates with reward 1; action 1 visits a second state, where actions 0 and 1 terminate with rewards 0 and 1 respectively. The four complete policies have exact return table

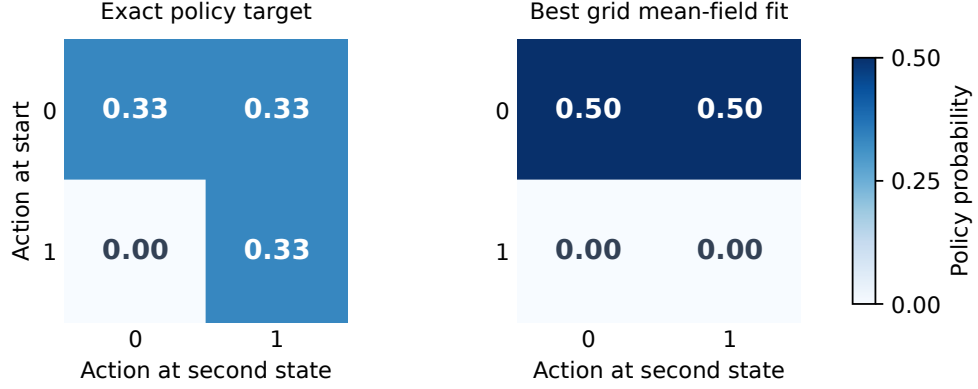
$$(J(a_0, a_1))_{a_0, a_1 \in \{0,1\}} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

At $\beta = 64$ with a uniform reference, the exact target is essentially uniform over the three reward-one policies. A product distribution cannot put equal mass on those three policies while excluding the fourth.

A grid with 2,001 points per Bernoulli coordinate gives reverse KL 0.405465 nats and a certified global family-gap interval $[0.368818, 0.405465]$. To obtain the certificate, let δ be half the grid spacing and $R = \max_{\pi} J(\pi) - \min_{\pi} J(\pi)$. Rounding each coordinate to its nearest grid point changes product-distribution total variation by at most 2δ , and changes each binary entropy by at most $h(\delta)$, where h is binary entropy in nats. Therefore the ELBO changes by at most

$$\epsilon = 2\beta R\delta + 2h(\delta).$$

High return does not certify accurate inference



Expected return is approximately 1 for both distributions.

Certified minimum family KL gap: 0.369 to 0.405 nats (beta = 64).

Figure 5.1: Exact four-policy diagnostic at $\beta = 64$. Both distributions achieve expected return essentially 1, yet the factorized family has a strictly positive certified reverse-KL gap. The displayed fit is a global grid search over Bernoulli coordinates, not a trained GridWorld policy or a claim about optimization convergence. Boundary probabilities are limits of the softmax family.

Subtracting ϵ from the best grid KL gives a lower bound on the global minimum over the family, clipped at zero. The grid KL is an upper bound; continuity extends the argument to boundary limits. This diagnostic therefore exhibits family error without attributing it to an optimizer or to poor control return (Figure 5.1). It is an illustrative finite MDP, not another large-scale benchmark.

5.6 Revisits can favor either execution rule

A second exact example has one decision state and horizon 3. Action 0 terminates with reward 1; action 1 returns to the same state with reward 0.4. If action 1 has probability p , committing to a sampled table gives value $1 + 0.2p$, while resampling on each visit gives $1 + 0.4p + 0.4p^2 - 0.6p^3$. At $p = 0.75$ these values are 1.15 and 1.271875. Negating both rewards reverses the ranking. Direct checks against the exact MDP evaluators reproduce both signs. Thus consistency is a distinct decision rule, not a universal control advantage.

5.7 Alternative sampling engines

The remainder of this chapter documents target-invariant MCMC and fixed-bank SMC alternatives. Their exactness and mixing conditions are not properties of the variational optimizer above. Negative sampling results remain useful evidence about the computational structure of policy inference.

5.8 Exact and sample-average policy scores

The sampler uses a score $\tilde{J}(\pi)$. With an explicit finite transition model, $\tilde{J} = J_H$ is evaluated exactly by dynamic programming. This variant targets Equation (3.1) exactly, but is model-based.

With simulator-only access, draw an immutable bank of K random-number

tapes $\Xi_K = (\xi_1, \dots, \xi_K)$ and define

$$\hat{J}_K(\pi; \Xi_K) = \frac{1}{K} \sum_{k=1}^K G_H(\pi, \xi_k). \quad (5.7)$$

Conditional on that bank, the implemented target is

$$P_{\beta,K}(d\pi \mid \Xi_K) \propto \exp\left(\beta \hat{J}_K(\pi; \Xi_K)\right) \mu(d\pi). \quad (5.8)$$

The same tape bank is reused for every policy evaluation in a chain. This is both common-random-number variance reduction and a definition of the finite-sample target. Refreshing the bank inside an ordinary Metropolis–Hastings ratio would generally invalidate detailed balance. For finite policy spaces and bounded returns, $\hat{J}_K$ converges uniformly to J , so Equation (5.8) converges to Equation (3.1); finite K is nevertheless an approximation, not an unbiased exact-target construction.

5.9 A learned proposal that cannot change the target

PPO-Clip [6] is trained first and then frozen. Its statewise action probabilities guide local moves, but PPO is neither the target nor an uncorrected component of the final sampler. Let $p_\phi(a \mid s)$ be the frozen probabilities and define

$$g_{\eta,\phi}(a \mid s) = \eta p_\phi(a \mid s) + (1 - \eta) \frac{1}{|\mathcal{A}|}, \quad 0 \leq \eta < 1. \quad (5.9)$$

The uniform component gives every action positive probability. If the current action is a , a proposed different action a' has probability

$$q_s(a' | a) = \frac{g_{\eta,\phi}(a' | s)}{1 - g_{\eta,\phi}(a | s)}, \quad a' \neq a. \quad (5.10)$$

Both the guide snapshot and η are stored with the run. Setting $\eta = 0$ gives the uniform-proposal ablation.

5.10 Single-site policy Metropolis–Hastings

One local proposal selects a decision state uniformly, changes only its action according to Equation (5.10), and leaves all other policy entries fixed. At inverse temperature b , accept π' from π with probability

$$\alpha_b(\pi, \pi') = \min \left\{ 1, \exp \left[b \{ \tilde{J}(\pi') - \tilde{J}(\pi) \} \right] \frac{\mu(\pi') q_s(a | a')}{\mu(\pi) q_s(a' | a)} \right\}. \quad (5.11)$$

The state-selection terms cancel. The experiments use uniform μ , so its ratio is one. The remaining Hastings ratio is essential because forcing a different action makes even a fixed categorical guide asymmetric. With full support, this kernel is irreducible on the finite policy space and satisfies detailed balance with the chosen exact or fixed-tape target.

5.11 Exact whole-policy independence refresh

Single-site edits can be ineffective on a sparse-return plateau, where many policies have the same score but reaching a useful policy requires several coordinated action changes. Each active replica update therefore chooses the single-site kernel with probability $1 - \rho$, or an independence proposal with probability ρ . The fixed whole-policy proposal is

$$Q_\phi(\pi) = w_{\text{map}} \mathbf{1}\{\pi = \pi_\phi^{\text{MAP}}\} + w_{\text{guide}} \prod_{s \in \mathcal{D}} p_\phi(\pi(s) \mid s) + w_{\text{uniform}} |\mathcal{A}|^{-|\mathcal{D}|}, \quad (5.12)$$

where $\mathcal{D}$ is the decision-state set and π_ϕ^{MAP} is the statewise mode of the frozen PPO guide. The defaults are $\rho = 0.10$ and $(w_{\text{map}}, w_{\text{guide}}, w_{\text{uniform}}) = (0.25, 0.70, 0.05)$. The components overlap, so the proposal density is the sum in Equation (5.12), evaluated by log-sum-exp, not merely the density of the component that generated the draw. The positive uniform component guarantees full support.

At inverse temperature b , an independent draw $\pi' \sim Q_\phi$ is accepted with

$$\begin{aligned} \log R_b(\pi, \pi') &= b\{\tilde{J}(\pi') - \tilde{J}(\pi)\} \\ &\quad + \log \mu(\pi') - \log \mu(\pi) \\ &\quad + \log Q_\phi(\pi) - \log Q_\phi(\pi'), \end{aligned} \quad (5.13)$$

$$\alpha_b^{\text{global}}(\pi, \pi') = \min\{1, \exp[\log R_b(\pi, \pi')]\}. \quad (5.14)$$

This is an ordinary independence Metropolis–Hastings step [7]. The complete proposal ratio is mandatory. Consequently, the guide, its mode, and the

branch probability affect efficiency but cannot change the invariant target. Full support gives an asymptotic finite-space convergence guarantee, not evidence that a finite run has mixed.

Forced-different proposals can be periodic in a two-action domain. We therefore mix every complete within-replica-plus-swap sweep with the identity kernel:

$$K_\delta = \delta I + (1 - \delta)K, \quad 0 < \delta < 1. \quad (5.15)$$

The stay-put decision is independent of the chain state, so it preserves the product target while giving every augmented chain state a positive self-transition. The default is $\delta = 0.05$; swap parity advances only on active sweeps. The branch settings, local and global proposal/acceptance counts, a separate count of accepted policy-changing global moves, δ , and the realized lazy-sweep count are stored with each run.

5.12 Fixed structural blocks and connectivity

The implementation also supports two separately selected branches: a path-only update and an ordered sweep over a fixed catalog of blocks. The catalog follows the modal guide from the start to a terminal state and then partitions its one-hop recovery neighbors into pairs (with a possible final singleton). This construction queries exact successor probabilities and is therefore model-assisted even when the scores use fixed tapes. A cycling modal path is rejected, rather than silently selecting a different construction from production samples. The path block uses a MAP/guide/uniform independence mixture restricted

to its coordinates. Its Hastings ratio sums all overlapping mixture components. For a small recovery block, let m denote the guide-MAP assignment and let R be the guide/uniform mixture conditioned on not choosing m . The proposal moves from any $x \neq m$ to m deterministically, and from m to $y \sim R$. Its reverse-to-forward proposal ratio is $R(x)$ in the first case and $1/R(y)$ in the second. Applying the corresponding Metropolis–Hastings correction preserves the conditional policy target. Composing the fixed block kernels preserves invariance, although the complete ordered sweep need not be reversible.

Invariance alone does not establish connectivity. If local and whole-policy branches are disabled, the active catalog must cover every decision coordinate. Even full coverage can leave a deterministic ordered sweep trapped in a parity class on a flat two-action problem. To prevent this, each scored local or structural substep uses

$$\alpha_{\text{substep}}(x, y) = 0.95 \min\{1, \exp[L(x, y)]\}.$$

This is a mixture of the ordinary MH kernel with an independent identity step. It preserves detailed balance and permits individual replicas and blocks to remain fixed while others move. Whole-sweep laziness alone cannot provide this property. Rejected substeps still count their proposal and score evaluation, so the acceptance accounting remains explicit.

The structural kernels have passed finite-policy implementation checks, but their medium-scale development runs have not established useful mixing. They are recorded experimental variants, not a validated scalability result.

5.13 Replica exchange across inverse temperatures

Single-site moves can mix slowly when good complete policies occupy separated modes. Replica exchange, also called parallel tempering, couples chains at inverse temperatures

$$0 = b_0 < b_1 < \cdots < b_{M-1} = \beta, \quad b_\ell^{(0)} = \beta\{\ell/(M-1)\}^p, \quad (5.16)$$

where $p > 0$ controls the initial ladder spacing. Development-only pilots may replace this power ladder with a fixed, explicitly listed ladder that places more replicas at measured exchange bottlenecks, following the feedback principle of Katzgraber et al. [8]. Adaptation stops before target samples are collected, and the endpoints remain zero and β . Replica ℓ applies the fixed mixture of Equations (5.11) and (5.14) with $b = b_\ell$. At a declared interval, four checkerboard swap sub-sweeps are applied by default, with parity alternating globally. These extra swaps reuse cached values and compose invariant kernels, so they can improve temperature-space flow without extra policy evaluations [9]. For replicas $i < j$, the swap acceptance probability is

$$\alpha_{i,j}^{\text{swap}} = \min\left\{1, \exp\left[(b_j - b_i)\{\tilde{J}(\pi_i) - \tilde{J}(\pi_j)\}\right]\right\}. \quad (5.17)$$

This is a Metropolis update for the product of tempered targets

$$\prod_{\ell=0}^{M-1} \mu(\pi_\ell) \exp(b_\ell \tilde{J}(\pi_\ell)),$$

and hence preserves the desired cold marginal at $b_{M-1} = \beta$ [10]. The PPO guide affects only proposal efficiency; Equations (5.11), (5.14), and (5.17) retain the target.

By default, all replicas start at the PPO statewise mode, while the $b_0 = 0$ chain is allowed to forget that initializer during burn-in. A separate overdispersed diagnostic initializes hotter replicas from the policy prior. Initialization cannot alter the stationary distribution, but it can strongly affect a finite run. Only thinned post-burn-in draws from the b_{M-1} replica are reported as target samples.

5.14 Diagnostics and cost accounting

Each run records the beta ladder, local and global proposals and acceptances, accepted policy-changing global moves, and adjacent swap proposals and acceptances for both the complete and post-burn windows, retained returns, return effective sample size, policy-value evaluations, lazy sweeps, simulator transitions, PPO training transitions, wall-clock time, and software and source hashes. Return effective sample size is necessary but can miss movement between policies with similar value. We therefore also record conservative and quantile summaries of action-indicator effective sample sizes, statewise switch-rate summaries, distinct sampled policies, walker temperature visits, endpoint transitions, round trips, and the number of walkers reaching the cold replica. Independent warm and overdispersed chains provide the between-chain check. Calibration uses two distinct decisions. A short two-chain probe tests only

whether the proposed ladder has viable post-burn edge exchange and endpoint flow; passing it merely permits the longer run. The full multichain gate then tests return $\widehat{R}$ and ESS, edge exchange, cold-reaching walkers and round trips, coordinate total variation, unique-policy fraction, mean policy Hamming movement, and switching at nondegenerate coordinates. Only the latter supports a mixing claim. A high-return cold replica cannot compensate for a failure at either stage.

For M replicas and scored proposal counts N_{local} , N_{global} , N_{path} , and N_{block} , summed over every replica and block repetition, logical policy-value evaluations are

$$M + N_{\text{local}} + N_{\text{global}} + N_{\text{path}} + N_{\text{block}}.$$

An ordered structural sweep can perform multiple evaluations per replica. Exact repeated-policy cache hits and actual score computations are logged separately and sum to this logical total. Swaps reuse current scores.

PPO training is part of the proposed method’s cost. Matched-budget comparisons therefore include every interaction and optimizer update used to train the guide plus all downstream policy evaluations. Exact dynamic-programming evaluations are logged separately because they consume model computations but zero simulator transitions.

5.15 Complementary estimators

Direct variational inference is defined at the start of this chapter in Equations (5.2)–(5.5). As a separate alternative, replicated policy SMC instantiates

each state action once in a lazy policy memo and advances K simulator replicas under that same policy. For newly exposed states U_t , its incremental log weight is

$$\Delta \log w_t = \frac{\beta}{K} \sum_{k=1}^K r_{t,k} + \sum_{s \in U_t} [\log \mu(a_s \mid s, m_{<s}) - \log q(a_s \mid s, m_{<s})]. \quad (5.18)$$

The mean reward is formed before exponentiation, proposal correction is applied once per newly assigned policy entry, revisits reuse the memo, and resampling copies the entire ancestry-dependent state. Sequential Monte Carlo can define powerful variational families [11] and particle objectives [12], but those results do not by themselves make an exponentiated noisy return an unbiased estimator of Equation (3.1). Here replicated SMC is evaluated against the explicitly stated fixed-tape target and against exact enumeration. Diagnostic controls include unguided and PPO-guided MCMC/SMC and PPO alone. All simulator interactions are counted.

Chapter 6

Experimental Design

The empirical study separates four questions: (i) does the implementation estimate the stated objective correctly, (ii) what approximation or optimization error remains, (iii) how well does the learned distribution control the environment under each execution rule, and (iv) what simulator and compute cost is required? Direct variational learning is the practical candidate. Sampling alternatives additionally require target-fidelity and mixing diagnostics; their failures are not silently reinterpreted as successful inference.

6.1 Domains and scope

Stochastic GridWorld is the planned primary study. Easy, medium, and hard families use, respectively, 3×3 , 5×5 , and 8×8 grids. Wall counts are 1, 4, and 13; swamp counts are 1, 3, and 9; gravel and dirt counts are each 1, 3, and 9. Relative to all grid cells, wall fractions are 11.1%, 16.0%, and 20.3%,

and each of the other negative terrain fractions is 11.1%, 12.0%, and 14.1%. Action success probabilities are 0.90, 0.85, and 0.80, with equal left/right slip; horizons are 20, 40, and 80. Frozen-map audits record reachability, shortest-path length and count, and single-cell start-goal bottlenecks. Map zero is used for development calibration; maps one and two are reserved for fresh-map validation after configuration selection. The confirmatory scientific design was frozen before generating ten fresh base geometries per tier. Each base is paired with its meadow representation and three fresh training blocks, yielding 30 independent base geometries and 90 paired geometry/training blocks. Maps are accepted only for the declared terrain counts, safe connectivity, and absence of prior or new spatial duplicates, never according to any policy’s performance. Small diagnostic instances retain exact enumeration; larger instances retain exact dynamic-programming evaluation of any fixed policy even when enumerating all policies is impossible.

Rewards accrue on arrival: pavement 0, gravel -1 , dirt -0.5 , meadow $+0.5$, goal $+5$, and swamp -5 , with an additional -0.1 per active transition. The goal and swamp absorb; their padded suffix earns zero. A blocked move stays in place and pays the actual destination terrain again. The paired reward landscape adds 1, 2, and 3 nonterminal meadow cells without changing the base geometry. Repeated meadow rewards are intentional: goal rate is not a utility selection criterion in this secondary panel. All seven cell types are therefore covered, without pooling navigation and meadow returns on an incompatible scale.

Only domains that add a distinct scientific stress test are considered after

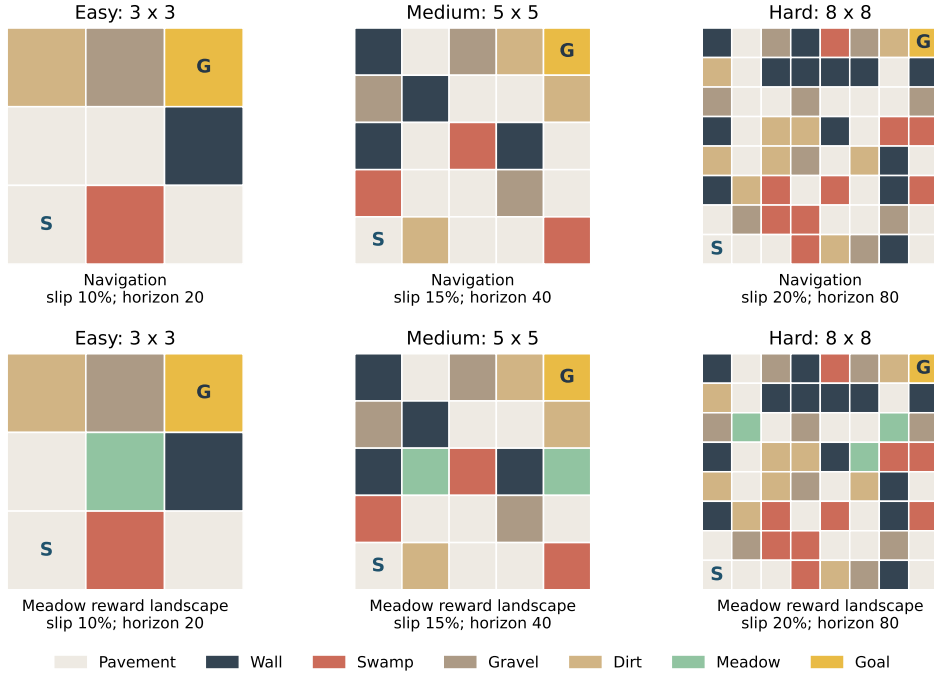


Figure 6.1: Illustrative development layouts for the three GridWorld tiers.

Navigation (top) and meadow reward landscapes (bottom) share base geometry, so they are not independent map replicates. S marks the start and

G the absorbing goal. These layouts illustrate tasks, not experimental performance. Protected layouts remain unused during method development and tuning.

GridWorld passes its medium gates:

A Tireworld extension would use a verified factored-state adapter; pyRDDL-Gym is a possible RDDL-to-simulator route [16]. Transition, termination, reward, and action-mask tests on a small instance are mandatory before any such results are included.

Table 6.1: Domain-extension decision. Extensions are conditional on the primary GridWorld study rather than required to enlarge the result table.

Domain	Added evidence	Decision
Blackjack	Compact nonspatial stochasticity and intrinsic return variance with two actions [13]	First low-cost extension after GridWorld
Tireworld	Stochastic failures, resources, sparse success, and competing routes [14]	Best second headline domain if time and compute remain
Taxi	Action masks and subgoals, but mostly deterministic and spatial	Optional smoke test only
Academic Advising	Factored multivalued state and concurrent actions [15]	Future structured-policy work
Continuous control	Requires a new reference measure and policy-space sampler	Out of scope

6.2 Competitors and controls

The competitor implementations preserve each published core update while using a common tabular code path for identical environment semantics, seeding, transition accounting, and artifacts. They are named by the exact implemented variant rather than by a broader software package.

The exact control is nonstationary and may use the known model, so it is an upper reference rather than a fair learned competitor. All learned and inferred output policies use stationary physical-state inputs. For correct finite-horizon backup targets, discrete SAC uses time-indexed twin critics and replay tuples, while Double Q-learning uses time-indexed tables and projects its result into the declared stationary policy class using state-time visitation.

Comparisons are stratified by information access. In the known-model stratum, exact-value tempering is tested for target fidelity and mixing against

Table 6.2: Learned competitors. Uniform random behavior and exact finite-horizon dynamic programming are controls, not competitors.

Reported method	Implemented variant	Role
REINFORCE	Monte Carlo policy gradient with learned state-value baseline [5]	On-policy score-function baseline
PPO-Clip	Clipped surrogate, Monte Carlo advantages, entropy bonus; no GAE or value clipping [6]	On-policy clipped-surrogate baseline and separately costed proposal trainer
Discrete SAC	Categorical actor, twin critics, target critics, replay, and fixed entropy temperature [17]	Off-policy entropy-regularized baseline
Categorical CEM	Distribution search over complete stationary deterministic policies [18]	Direct black-box policy search
Double Q-learning	Online tabular Double Q-learning with linearly annealed epsilon-greedy exploration [19]	Value-learning baseline

exact enumeration or known-model controls. Fixed-tape scoring alone does not imply simulator-only access: the current path/recovery block builder reads exact successor probabilities. Such runs remain model-assisted even when scores come from rollouts. In a simulator-only stratum, that builder must be disabled or replaced by separately counted simulator-based construction. Eligible fixed-tape variants, direct variational learning with its PPO initializer, and the five learned competitors receive generative-simulator access. Exact-model tempering may be displayed beside simulator-only learners as a descriptive reference, but such a cross-stratum table cannot support a sample-efficiency superiority claim. Model value evaluations, simulator transitions, and guide training transitions are therefore reported in separate cost columns.

6.3 Common evaluation semantics

For the committed-policy diagnostic, the evaluator samples a complete deterministic policy from the reported distribution and commits to it for the entire episode. For an explicit particle or MCMC sample, that complete policy is used directly. For statewise categorical outputs, one action is sampled once at every decision state before evaluation. The fixed table’s expected return and goal probability are evaluated exactly; their distributional means are estimated with 2,048 independent table draws at the terminal checkpoint. The original 64-draw development learning checkpoints remain unchanged and their first 64 terminal draws are preserved in the higher-precision attachment. For independent VI or baseline policy draws, the sample standard deviation describes policy variation and the standard error describes conditional evaluation precision. Neither measures uncertainty across training runs. Correlated MCMC draws or resampled particles require separate dependence and ancestry diagnostics; the independent-draw formula is not valid for them. This rule is an intervention on stochastic baseline outputs, not their native deployment semantics. Practical comparisons and selection use method-native expected original return: commitment for the inference distributions and CEM; per-visit action draws for PPO, REINFORCE, SAC, and stochastic tie-breaking in Double Q. Committed-policy values of stochastic baselines answer a narrower policy-class question and cannot replace their native result. A sampler’s statewise marginals cannot replace its committed-policy result because they remove policy-level dependence. The statewise maximum-probability policy is also reported separately. Secondary metrics include

actual episode-return lower tails (not a histogram of policy expected returns), separately counted evaluation interactions, distributional target error where enumeration is possible, and mode recovery.

6.4 Fair budgets and tuning

Every comparison uses the same maps, training seeds, evaluation streams, horizon, reward, and termination convention. Current GridWorld runs use $\gamma = 1$; discounted baseline semantics must be validated before any $\gamma < 1$ experiment. Method-specific tuning grids are allowed because a single global learning rate is not a fair tuning policy. Each tunable entry receives its canonical setting and two prospective single-parameter alternatives on all six panel/tier strata at 100k transitions. Actual allocated CPU costs are disclosed and bounded, not assumed equal between learners. Exactly matching canonical runs may be reused, with their original compute retained in the ledger. Tuning maps and seeds are disjoint from the final confirmatory set.

The inverse temperature defines the target and is not selected after observing held-out control performance. Exact diagnostic maps use a predeclared sensitivity grid; the headline GridWorld target freezes one value on map-zero development data. The prospective pure-inference screen contrasts $\beta = 64$ with $\beta = 16$, or changes policy batch size from 32 to 8 without changing temperature. Initialized inference contrasts guide fractions 0.5 and 0.75, or the same batch-size change. One configuration per entry is selected over equally weighted tier/panel normalized return gaps, not one winner per maze. Every

configuration gets the same 2,048-table terminal readout before selection. This is a bounded target/configuration choice on development data, never a best-of-grid choice using protected performance.

All interactions used by PPO guide training, variational refinement, SMC, policy-bank evaluation, and sample-average MCMC count toward the proposed method. Exact policy-value evaluations, optimizer updates, wall time, CPU or accelerator time, and peak memory are reported separately. A generally applicable exploration aid is offered to compatible methods or labeled as a separate assisted ablation. Competitor bugs are fixed; no baseline is weakened to improve a comparison.

New variational and baseline runs enable `strict_transition_budget`: the declared allowance is a maximum for all training transitions, including the initializer. Refinement receives the total minus actual guide usage. Complete-return reservation can leave a small unused allowance, which is reported. Historical tempering configurations retain different semantics: their `transition_budget` covers guide training, and subsequent sampling adds cost. They are not relabeled as strict-cap comparisons. Full-budget PPO under the same allowance is an essential attribution control; the current control starts from scratch, rather than resuming the guide optimizer.

6.5 Cost-effective execution order

After the claim-specific checks in Section 4, small technical runs verify every learner’s artifacts, budgets, and environment semantics. A paired budget-

response check informs resource requests without dropping weak methods. The bounded 100k screen precedes 200k validation on two fresh base geometries per tier, two independent training seeds per geometry, and both terrain panels, retaining every declared competitor and control. The two panels share geometry and seed blocks; they do not double independent map replication. Configuration rankings may change between 100k and 200k.

Only after this validation are the inference engine, confirmatory size, cap, analysis, and exclusions frozen before protected outcomes. Running jobs are limited to two, at most one SAC job, using separate serial lanes. Independent numeric and scheduler checks precede phase expansion. Technical failures halt expansion; low returns do not justify excluding a method or layout. All attempts are preserved, and a retry requires an explicit new artifact identity rather than overwriting completed or partial results.

The frozen main engine is PPO-initialized variational inference; pure inference and the full-budget assisted PPO control remain in the comparison. Every learned run receives the same 200k total active-interaction ceiling and a 2,048-table terminal readout. A separate 4,096-episode native deployment evaluation records actual returns, goal arrival, swamp termination, and timeout. Random, return-optimal and goal-optimal controls are evaluated once per map/panel, not duplicated for every training seed. Sampling and independent scalar replay run on compute nodes; their actual interactions and allocated CPU costs are kept separate from learning costs.

The primary report preserves original expected returns for each of the six tier/panel strata, averaging training blocks within maps and maps equally.

The secondary composite gives equal weight to those six strata after normalization between each map’s random and return-oracle anchors; lower gap is better. Degenerate anchor gaps retain raw results and undefined normalized values, without clipping or redrawing. Joint hierarchical bootstrap resamples base geometries, then paired training blocks, together across methods and panels (10,000 draws, seed 0). Its 95% intervals are descriptive and marginal, not simultaneous tests or proof of equivalence. Episode-tail figures use the predeclared medium/navigation and hard/meadow categories and aggregate all their maps and training runs, rather than selecting attractive instances.

The same order applies to domain extension: one exact Blackjack instance, one seed per method, then three-seed timing, and only then a complete table. Tireworld work begins only after the GridWorld confirmatory manifest is frozen.

6.6 Current empirical status

The redesigned terrain confirmatory study remains incomplete; no pilot score is a final result. The numerical material below records HISTORICAL development diagnostics from the earlier larger-grid study (8, 16, and 32 cells per side). Those maps, budgets, and seed cohorts are different from the current 3/5/8 terrain design. They are retained for methodological context, not pooled into its primary tables or used as protected evidence. Final results in the thesis and compact paper will use the same complete independently verified data.

Table 6.3: Medium GridWorld development comparison: one map, training seed 572, and a 1.2M total-transition cap. Parentheses are Monte Carlo standard errors over 256 committed-policy draws, not uncertainty across training runs. Native means action resampling. Wall time includes offline evaluation.

Method	Committed (SE)	Native	Seconds
VI (PPO-initialized)	0.725 (0.044)	0.992	40.9
PPO + count bonus	0.656 (0.098)	1.015	109.9
PPO	0.008 (0.137)	0.766	103.9

6.6.1 Variational allocation and full-budget controls

A cached medium-map PPO guide trained for 200,015 transitions initialized three independent refinements per rollout setting. With approximately one million additional transitions, $K = 1$ produced committed returns 0.581, 0.595, and 0.548; $K = 4$ produced -0.144, -0.244, and -0.087. The corresponding update counts were 662–669 versus 157–160. Every refinement improved its paired guide estimate and estimated ELBO. Evaluation used 512 independent policy draws, coupled within each guide/refinement pair. These refinements share one guide and map, so they do not establish robustness across independently trained guides.

Fresh-guide, strict-cap runs then used nominally 200k guide transitions and 1M refinement transitions. On the easy map, committed value rose from the guide’s 3.010 to 3.318, below the nonstationary upper control 3.451. Actual usage was 1,199,880 transitions. The medium run used 1,199,725 transitions and achieved committed value 0.725 versus its guide’s -0.225. Table 6.3 includes full-budget PPO controls at the same medium training seed 572. The VI guide used count-bonus coefficient 1.0; refinement used original rewards.

Assisted PPO used coefficient 1.0 throughout, and standard PPO used zero. Both controls trained from scratch.

VI’s committed value is slightly above assisted PPO’s and its native value is slightly below. The committed difference 0.069 is smaller than the combined conditional evaluation standard error. Neither equivalence nor superiority is established. The shorter local runtime is likewise one observation, not a general speedup claim. These runs share executable snapshot and map; their wall times include offline exact evaluation and use one numerical-library thread.

6.6.2 Hard-map exploration and execution semantics

Figure 6.2 displays the same completed development runs under both execution rules. The hard-panel allocation change is separately labeled; it is not an increase in the total simulator allowance.

The unchanged 200k-guide allocation failed on hard map zero with seed 573: all guide and refinement batches returned the no-goal value -38.4. The estimated ELBO rose through reduced KL, without improved control. The map is not unreachable: its finite-horizon upper return control is -4.001.

A separately frozen two-step diagnostic first trained full-budget assisted PPO. Its first successful batch occurred near 538k transitions, beyond the original guide allowance. Native goal probability reached 0.995. This satisfied the predeclared condition for testing a 1M-guide/200k-refinement reallocation at the same total cap, map, and seed. That run consumed 999,721 guide

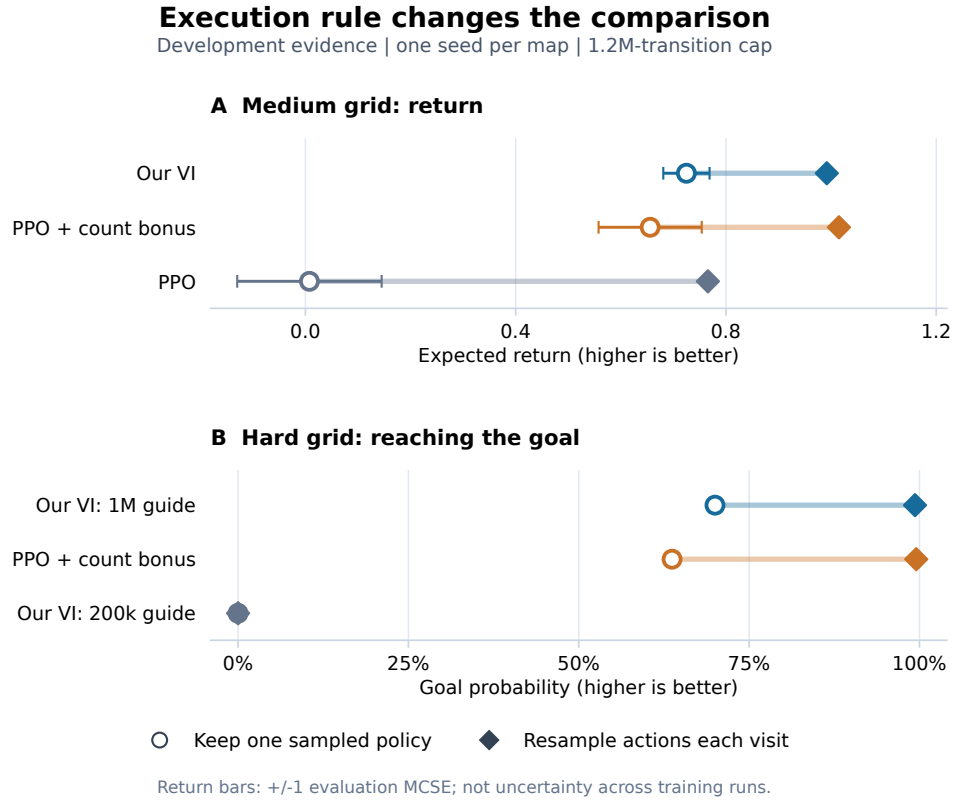


Figure 6.2: Execution-aware development comparison with a common 1.2M-transition cap. The medium panel uses seed 572 and the hard panel seed 573, each on map zero. Open circles commit to one sampled policy; filled diamonds resample actions on each visit. The return error bars are one conditional evaluation standard error over 256 policy draws, not training-level uncertainty. The hard-panel guide allocations retain the same total allowance. PPO with the count bonus is an explicitly assisted control; standard PPO is separate. These single-seed results motivate, but do not replace, multi-seed comparison.

Table 6.4: Completed BGU development calibration, with a strict 1.2M-transition cap. Rows report means over n independent training seeds. Subscripts c/n mean committed-table/native action-resampling execution; S is goal probability in percent. Time includes offline evaluation. Map 100 is a separate development check, not held-out confirmation. PPO uses count bonus 1 throughout; VI uses it only for initialization.

Grid/map	Method	n	J_c	J_n	S_c	S_n	Seconds
Medium/0	VI	3	-4.973	-4.695	65.9	66.8	40.5
Medium/0	PPO + bonus	3	0.449	0.970	98.5	100.0	130.4
Hard/0	VI	3	-17.597	-6.884	74.1	99.4	170.0
Hard/0	PPO + bonus	3	-15.493	-6.076	78.0	99.7	174.6
Medium/100	VI	1	0.418	0.906	99.0	100.0	42.0
Medium/100	PPO + bonus	1	0.781	1.048	99.4	100.0	129.9
Hard/100	VI	1	-13.535	-7.084	88.0	99.8	173.9
Hard/100	PPO + bonus	1	-11.934	-6.744	89.3	99.7	171.5

and 199,549 refinement transitions. It achieved committed return -19.219 (conditional MCSE 0.679), compared with its guide’s -23.077. Native return was -6.775.

Committed goal probability reached 0.700, versus zero with the short guide; native action resampling reached 0.993. Full-budget assisted PPO achieved committed return -20.900 and native return -6.415, with goal probabilities 0.637 and 0.995 respectively. Wall times were 132.4 seconds for reallocated VI and 133.9 seconds for assisted PPO, including evaluation. This single-seed allocation diagnostic supports testing independent guide seeds, not a general hard-grid success or superiority claim. The large execution gap is also a counterexample to treating policy commitment itself as a control advantage.

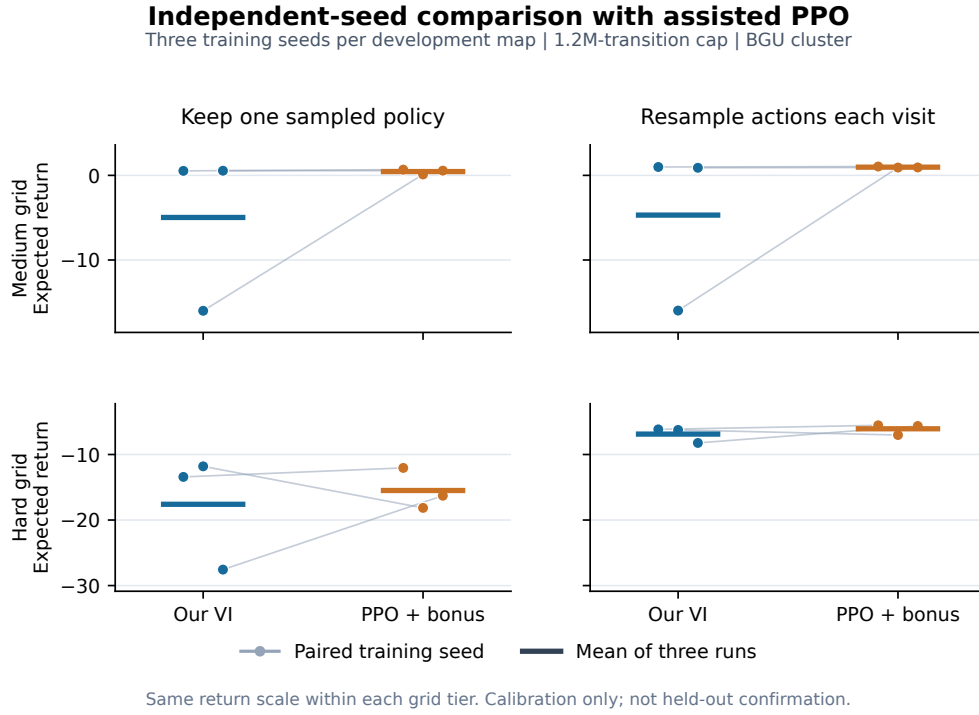


Figure 6.3: Independent-guide calibration on medium and hard development map zero. Each point is one complete training run; lines pair seeds 601–603, and horizontal bars are arithmetic means. Both methods have a strict 1.2M-transition cap. VI uses nominally 200k guide transitions on medium and 1M on hard; PPO uses the exploration bonus throughout. The failed medium VI run is retained. With only three seeds per map, these are descriptive results, not equivalence or superiority tests.

6.6.3 Independent-guide BGU calibration

The next frozen screen completed all 16 runs on BGU compute nodes. Training seeds 601–603 were used on medium and hard development map zero, and seed 604 on separately reserved development map 100 in each tier. Held-out maps 1–3 remained unused. The total cap was 1.2M transitions per run, with nominal VI guide allocations of 200k on medium and 1M on hard. The target, learning rate, count bonus, and $N = 32$, $K = 1$ settings were fixed before dispatch. PPO used count bonus 1 throughout; VI used it only in initialization and unshaped rewards during refinement. These are explicitly assisted comparisons.

Every artifact passed checks of configuration, map hash, executable snapshot, strict budget, finite oracle-bounded metrics, and Slurm provenance. The cluster cohort shares its software environment and is not pooled with older local timing or seed results. Evaluation uses 256 independent complete-policy draws; their conditional Monte Carlo precision is not uncertainty across training runs. The plotted arithmetic means in Figure 6.3 retain every seed, including the failed medium run. Table 6.4 also reports the second-map checks separately rather than blending maps.

PPO has higher committed and native return in seven of the eight paired map–seed comparisons. Primary-map mean committed returns are -4.973 versus 0.449 on medium and -17.597 versus -15.493 on hard (VI versus PPO). The corresponding native means are -4.695 versus 0.970 and -6.884 versus -6.076. The second development maps also favor PPO in return under both rules. This does not establish a universal ranking, but it contradicts extrapolating

the earlier favorable single-seed pilot into a robustness claim.

Medium seed 602 isolates the failure. Its guide stops at 199,923 transitions with native return -15.936 and committed return -15.999. Every one of its 196 refinement minibatches returns -16 with essentially zero between-policy return variation. The estimated ELBO rises from -1140.570 to -1026.622 through the entropy term, not successful navigation. Final native goal probability is 0.00371; committed goal probability is approximately 3.1×10^{-6} . Continued assisted PPO with the same seed reaches its first positive batch mean at 318,132 transitions. Some guide batches had successful trajectories earlier, so first-ever success is not by itself a sufficient handoff criterion.

The primary medium-map mean wall times are 40.5 seconds for VI and 130.4 for PPO; hard-map means are 170.0 and 174.6 seconds. They include offline evaluation and describe this CPU implementation, not hardware-independent algorithmic complexity. A shorter failed run is not a performance advantage. A separate hard-grid SAC throughput screen consumed 4,992 transitions in 325.56 seconds; its larger conditional profile was therefore not released. That tiny-budget profile is cost evidence only, not a fair SAC control-performance comparison.

6.6.4 Bounded allocation follow-up

A separately frozen follow-up reallocated the medium allowance to 600k guide and the remainder refinement, without changing either algorithm’s updates or increasing total interactions. Selected failure seed 602 recovered to committed goal probability 0.9863 and native probability 0.9996. This

Table 6.5: Separate medium-grid allocation follow-up. VI uses a 600k nominal guide within the unchanged 1.2M total cap. Seeds 606/607 are fresh paired checks; seed 602 is a selected failure repair, and seed 604 uses development map 100. Neither selected case is pooled with the fresh-seed estimate. Subscripts c/n denote commitment/native resampling; goal probabilities S are percentages.

Role/seed	Method	J_c	J_n	S_c	S_n	Seconds
Repair/602	VI	0.369	0.862	98.6	100.0	112.2
Fresh/606	VI	0.628	0.975	99.1	100.0	95.2
Fresh/606	PPO + bonus	0.626	1.055	98.5	100.0	188.7
Fresh/607	VI	-0.004	0.737	97.8	100.0	109.8
Fresh/607	PPO + bonus	0.258	0.874	98.3	100.0	175.2
Map 100/604	VI	0.752	1.050	99.0	100.0	110.9

Longer initialization, unchanged total interaction budget

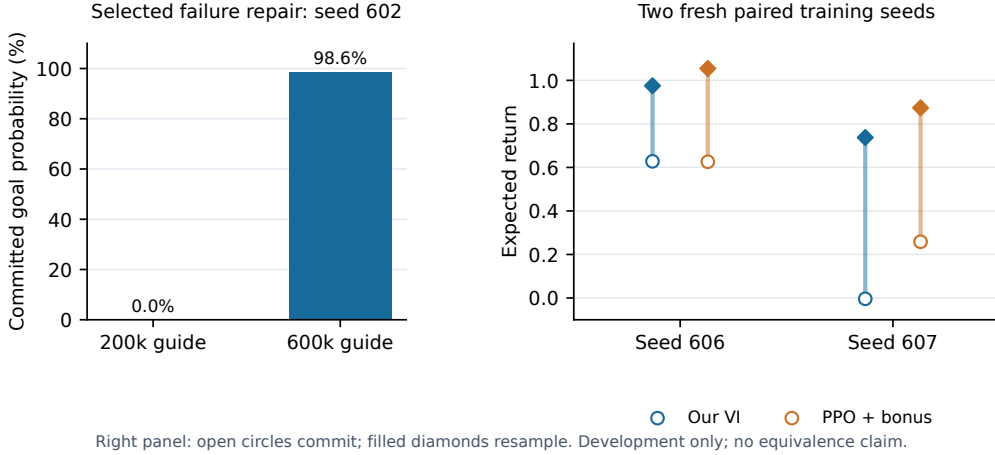


Figure 6.4: Separating repair from validation. Left: the originally failed seed 602 under two allocations of the same total budget; its selection was outcome-dependent. Right: two subsequently declared fresh seeds, showing both execution rules for VI and assisted PPO. The successful repair is not counted as an independent fresh seed. This small development screen does not establish practical equivalence.

satisfied the predeclared conditions for two fresh paired seeds, 606 and 607, and a second-map VI check using seed 604. The repair is not pooled into the fresh-seed estimate.

Both fresh VI runs avoid collapse, with committed goal probabilities 0.9910 and 0.9779; native probabilities exceed 0.9998 for both methods. Committed returns are 0.628 and -0.004 for VI versus 0.626 and 0.258 for PPO. Native returns are 0.975 and 0.737 versus 1.055 and 0.874. Thus the improvement removes the observed exploration failure but does not establish superiority: PPO still has higher native return on both fresh seeds. On development map 100, VI has committed/native returns 0.752/1.050 versus the existing PPO control's 0.781/1.048. These near values are descriptive, not an equivalence test.

All six completed follow-up artifacts passed the integrity audit. Two PPO attempts were interrupted by an unhandled Slurm warning signal; the launcher was corrected and only those attempts were repeated from scratch at the same five-minute wall limit. Failed logs and partial artifacts are retained and their wasted compute is counted separately. No completed scientific outcome was replaced. Follow-up timings come from a different node than the original calibration; only the within-follow-up fresh pairs have matched hardware. Their mean wall times, including evaluation, are 102.5 seconds for VI and 182.0 for PPO. This small CPU comparison motivates a costed larger study, not a general speedup claim. Held-out maps remain unused.

6.6.5 Sampling alternatives: development findings

Preliminary medium-GridWorld exact-value tempering diagnostics produced high-return committed policies but exposed a sharp intermediate-temperature exchange bottleneck. Four independent 32-temperature, power-four chains, including an overdispersed chain, each retained 800 draws. They failed the predeclared gate: split return $\hat{R} = 1.096$, minimum per-chain return ESS of 10.5, minimum whole-run adjacent-swap acceptance of 0.049, no completed round trip, and maximum pairwise state-action total variation of 0.699. The principal warm-chain bottleneck lay between inverse temperatures 1.44 and 1.98. Because the legacy acceptance counter included burn-in, subsequent runs also store a separate post-burn rate; the ESS, round-trip, and between-chain failures do not depend on that counter. A subsequent two-chain, 4,000-iteration custom-ladder probe also failed. Its warm chain had post-burn acceptance between 0.598 and 0.997 on every edge, but no walker traversed both endpoints. In the overdispersed chain, replicas below the cold endpoint remained on a flat random-policy plateau near return -16 , while the cold replica remained near return 1.1; the final edge consequently accepted no swap. High acceptance inside the bad plateau therefore did not indicate useful mixing.

Subsequent whole-policy and structural-block rescue attempts did not establish adequate mixing. An isolated path-proposal screen used four independent 4,500-step chains at $\beta = 3.42$, discarding 1,200 steps and retaining every tenth subsequent state. Low-initialized chains remained near return -16 ; high-initialized chain medians were approximately -1.56 and -1.41 . Path

Table 6.6: Required confirmatory reporting. No entry is populated from pilot runs.

Question	Comparison	Primary report
Estimator correctness	exact differentiation	gradient error under stochastic visitation
Family error	enumerable posterior	variational gap and omitted dependence
Sampling alternatives	chains and ladders	ESS, swaps, ancestry, and agreement
Control quality	five competitors	native and committed return and goal probability
Robustness	map families and seeds	uncertainty and failed-run count
Scalability	matched accounting	interactions, time, and memory

banks fitted on one chain from each region failed to generate useful bidirectional crossings on the other chains. These measurements do not estimate the relative posterior mass of the regions.

A bounded three-seed guide-field replica-exchange screen subsequently found high-return policies but did not validate sampling adequacy. Return effective sample sizes were 28–66, split $\hat{R}$ was 1.032, maximum between-chain coordinate total variation was 0.578, and completed round trips were 0, 1, and 1. A fixed-bank SMC screen retained only 3, 4, and 4 root ancestors from 256 particles and returned values near -16. These alternatives have different costs and information access; they are not pooled into a simulator-efficiency leaderboard. Historical failed checks remain failed. The practical VI candidate instead requires family-error, learning-stability, and execution-aware control checks before full-scale confirmation.

Chapter 7

Discussion and Limitations

7.1 Relation to policy-gradient learning

The current practical method uses a score-function policy gradient, but its random object is a complete deterministic policy table and its regularizer acts on a distribution over those tables. The PPO guide initializes the variational distribution once. Subsequent updates change its logits directly using original rewards; fresh policies are sampled each iteration. Neither trajectory weights nor particle genealogy are carried across updates. The relation to REINFORCE is therefore explicit, not denied: the difference lies in the sampled object, objective, and execution semantics.

The separately implemented MCMC alternative behaves differently. Its guide is frozen while complete policies evolve through corrected proposals and replica swaps; there is no guide-gradient update during that chain. Consequently, claims about invariant distributions or MCMC mixing must not be used as

guarantees for the variational optimizer.

Determinism is a representation and execution choice, not a general computational advantage. It makes repeated actions consistent but does not remove stochastic transitions or the exponentially large policy space. On the hard development instance, action resampling gives much higher goal probability than commitment for both VI and PPO. A committed wrong action can persist at every revisit. This is a substantive behavioral difference and a limitation for control, not a result to hide by reporting only the more favorable execution rule.

7.2 Approximation and interpretation

With a uniform reference, modes of the target maximize expected return. A finite-temperature draw need not do so. The resulting dispersion depends on the reference and temperature; it is not calibrated epistemic uncertainty about an unknown environment.

The direct ELBO avoids exponentiating noisy return estimates, but its gradient can have high variance. Mean-field policies cannot express general cross-state dependence, and reverse-KL optimization may represent one policy region while omitting others. An unbiased estimator is a statement about the expected update at fixed parameters, not a guarantee of global optimization or accurate posterior coverage. Family error needs its own small-instance characterization and behavioral tests.

Replicated SMC retains policy ancestry under an explicitly fixed-bank approx-

imation but can degenerate severely. The tested medium configuration kept only a few root ancestors and almost never reached the goal. Completing unvisited table entries makes policies syntactically distinct without establishing useful behavioral diversity.

For MCMC, exact Hastings correction guarantees invariance, not finite-run mixing. Local edits and even global proposals can fail to traverse high-dimensional policy modes. Adjacent swap acceptance, return ESS, coordinate movement and agreement between dispersed chains must therefore be checked together. The tested bridge produced good policies but failed to establish full distributional agreement.

7.3 Exploration and guide attribution

Sparse rewards create a common exploration problem [20]. In the hard development run, a 200k-transition guide never achieved success, and all refinement returns were equal. The entropy term then improved the estimated ELBO without improving control. Full-budget assisted PPO found success only after that initial allowance. Reallocating the same total budget toward guide learning helped in a subsequent conditional development test. Repetition across independent guides is required before treating this allocation as a robust choice. The subsequent BGU calibration demonstrates why: one of three medium-map seeds also collapses after the 200k handoff. Successful trajectories occurred during its guide training, but the resulting distribution was still inadequate for committed-policy refinement. The other seeds do

not justify discarding this failure or claiming that initialization is solved. Increasing the medium guide to 600k transitions within the same total cap recovers that selected seed and avoids collapse in two fresh seeds. PPO still has higher native return on both fresh seeds. The follow-up narrows the exploration problem but does not resolve initialization robustness across maps or establish an inference advantage.

This dependence on a guide is part of the method’s cost and limitation. Improvement over an under-budget initializer does not establish improvement over full-budget PPO. Standard PPO and exploration-assisted PPO are separate controls. In the assisted comparison, PPO retains the count bonus throughout training, whereas VI uses it only during initialization; this distinction must remain visible. Performance against an unassisted baseline alone cannot isolate an inference contribution.

7.4 Access, policy class and scope

Direct variational refinement uses only simulator returns for learning. Exact dynamic programming is used offline for evaluation. In contrast, exact-score MCMC uses model information during inference, and the optional structural block builder reads exact successor probabilities even when its scores use rollout tapes. These are separate information-access strata, not interchangeable sample-efficiency competitors.

The fixed-bank target converges to the intended expected-return target under finite-policy bounded-return assumptions, but finite bank size remains an

approximation. Bounded returns also permit a nonnegative Poisson estimator for a pseudo-marginal construction; Appendix A gives its identity and variance. Its potentially prohibitive cost prevents mathematical exactness from serving as a practicality claim.

All reported output policies are stationary in physical state, while the finite-horizon dynamic-programming upper control is nonstationary. PPO uses Monte Carlo advantages without GAE or value clipping. Discrete SAC uses a fixed entropy temperature and time-indexed twin critics; Double Q-learning uses time-indexed tables before an occupancy-weighted projection to a stationary output. These precisely specified implementations preserve their core updates but do not claim every enhancement of the algorithm families. Current GridWorld studies use $\gamma = 1$.

Finally, the present development samples cannot establish generality or practical equivalence. Independent training seeds and held-out maps are the experimental units; repeated policy draws only improve conditional evaluation precision. A nonspatial Blackjack study could add evidence at low cost, whereas Tireworld would test structured stochastic failure with greater implementation effort. These extensions cannot replace completing the primary comparison and its uncertainty analysis.

Chapter 8

Conclusion

Expected-return policy inference is meaningful only after the policy space, reference measure, inverse temperature, and execution rule are explicit. Averaging environment randomness before exponentiation distinguishes its target from an exponential-trajectory objective. This established Gibbs construction motivates an implementation whose approximations can be tested independently of its control performance.

The practical candidate uses simulator-based variational learning over complete policy tables, initialized by a separately costed PPO guide. Its masked score-function estimator is unbiased under the stated sampling assumptions, and exact small-instance tests exercise the actual implementation. It does not follow that a finite run globally optimizes the objective or represents the full target. Corrected MCMC and fixed-bank SMC remain documented alternatives with separate invariance, approximation and mixing requirements. Development results identify useful questions rather than settle the final

comparison. One rollout per policy permits more effective refinement than four in the tested allocation. The independent-guide cluster calibration, however, includes a medium-map collapse, and assisted PPO has higher return in seven of eight paired comparisons. Hard-map failures and conditional budget reallocation expose the importance of exploration before refinement. A bounded medium-map reallocation recovers the selected failure and avoids collapse in two fresh seeds, while their native returns still favor PPO. The sharp difference between commitment and action resampling also prevents equating policy consistency with improved control.

Strict total simulator caps and explicit execution semantics now support a reproducible confirmatory study. The remaining evidence must characterize initialization robustness, larger-instance family error, held-out performance and full cost against the five declared baseline families. Competitive or mixed results can support a useful policy-inference contribution; neither universal superiority nor calibrated posterior uncertainty is claimed.

Appendix A

Core Derivations

Gibbs variational identity. Let P_β be the target in Equation (3.1). For any $Q \ll \mu$,

$$\text{KL}(Q \| P_\beta) = \mathbb{E}_Q \left[\log \frac{dQ}{d\mu} - \beta J_H(\pi) + \log Z_\beta \right] \quad (\text{A.1})$$

$$= \log Z_\beta - (\beta \mathbb{E}_Q[J_H(\pi)] - \text{KL}(Q \| \mu)). \quad (\text{A.2})$$

Non-negativity yields Equation (3.3), with equality at $Q = P_\beta$.

Why a rollout is valid in the direct objective. For $\pi \sim q_\theta$ and $\tau \sim p(\cdot | \pi)$, iterated expectation gives

$$\mathbb{E}_{\pi, \tau}[G_H(\tau)] = \mathbb{E}_{\pi \sim q_\theta}[J_H(\pi)].$$

No exponential or logarithm is applied between the inner and outer expectations. Applying the score-function identity to Equation (5.2), and dropping

terms whose expected score is zero, gives Equation (5.5).

Local detailed balance. At inverse temperature b , write

$$\tilde{P}_b(\pi) \propto \mu(\pi) \exp(b\tilde{J}(\pi)).$$

The Metropolis–Hastings ratio is

$$\frac{\tilde{P}_b(\pi')q(\pi \mid \pi')}{\tilde{P}_b(\pi)q(\pi' \mid \pi)} = \exp\left[b\{\tilde{J}(\pi') - \tilde{J}(\pi)\}\right] \frac{\mu(\pi')}{\mu(\pi)} \frac{q_s(a \mid a')}{q_s(a' \mid a)}.$$

The uniform decision-state selection cancels, producing Equation (5.11). Full support makes every one-coordinate change possible, and a finite sequence of such changes connects any two tabular policies.

Replica-swap ratio. Before a swap, the factors for replicas i and j are proportional to $\mu(\pi_i) \exp(b_i \tilde{J}(\pi_i)) \mu(\pi_j) \exp(b_j \tilde{J}(\pi_j))$. After exchanging the two policies, the reference-measure factors cancel and the new-to-old density ratio is

$$\exp\left[(b_j - b_i)\{\tilde{J}(\pi_i) - \tilde{J}(\pi_j)\}\right],$$

which gives Equation (5.17). Alternating disjoint adjacent pairs changes scheduling efficiency but not this invariant product distribution.

Fixed-tape convergence. For every fixed policy, the strong law gives $\hat{J}_K(\pi) \rightarrow J(\pi)$ almost surely. Because the tabular policy space is finite, convergence holds simultaneously for all policies. Continuity of exponentiation

and finite normalization then imply $P_{\beta,K}(\pi \mid \Xi_K) \rightarrow P_\beta(\pi)$ for every policy. This argument justifies a convergence study in K ; it does not make a finite tape bank exact for P_β .

A nonnegative unbiased policy weight from bounded rollouts. Suppose a simulator return satisfies $G_\pi \geq a$ almost surely, where the finite-horizon reward specification supplies the known constant a . For $\lambda > 0$, draw $N \sim \text{Poisson}(\lambda)$ and, conditional on N , independent returns $G_{\pi,1}, \dots, G_{\pi,N}$. Define

$$\widehat{W}_\lambda(\pi) = \exp(\beta a) \prod_{i=1}^N \left(1 + \frac{\beta(G_{\pi,i} - a)}{\lambda} \right). \quad (\text{A.3})$$

Every factor is nonnegative. Conditioning first on N , then using the probability generating function of a Poisson variable, gives

$$\mathbb{E}[\widehat{W}_\lambda(\pi)] = \exp(\beta a) \mathbb{E}_N \left[\left(1 + \frac{\beta(J(\pi) - a)}{\lambda} \right)^N \right] \quad (\text{A.4})$$

$$= \exp(\beta a) \exp(\beta(J(\pi) - a)) = \exp(\beta J(\pi)). \quad (\text{A.5})$$

Thus a pseudo-marginal Metropolis–Hastings chain on the policy and estimator randomness has the exact policy target as its marginal [21, 22]. The identity is not automatically an efficient algorithm. Its expected rollout count is λ , and direct calculation gives

$$\frac{\mathbb{E}[\widehat{W}_\lambda(\pi)^2]}{\exp(2\beta J(\pi))} - 1 = \exp\left(\frac{\beta^2 \mathbb{E}[(G_\pi - a)^2]}{\lambda}\right) - 1. \quad (\text{A.6})$$

Bounded relative variance requires rollout cost scaling with $\beta^2 \mathbb{E}[(G_\pi - a)^2]$, potentially prohibitive.

A.1 Exact one-decision diagnostic

Consider one state, one decision, and two policies. The safe action returns c surely; the variable action returns $+1$ or -1 with equal probability. There are no state revisits, exploration difficulties, resampling ancestry, or policy-representation differences. Set $\beta = 1$ and use a uniform prior. The expected-return target and exponential-trajectory target respectively give

$$P_{\text{mean}}(\text{variable}) = \frac{1}{e^c + 1}, \quad (\text{A.7})$$

$$P_{\text{trajectory}}(\text{variable}) = \frac{\cosh(1)}{e^c + \cosh(1)}. \quad (\text{A.8})$$

When $c = 0$, both policies have mean zero, so the first probability is $1/2$; the second is approximately 0.606776 . When $c = 0.2$, the safe policy has the higher mean, yet the trajectory target still assigns the variable policy approximately 0.558180 mass. The expected-return target instead assigns the safe policy approximately 0.549834 mass. Preference for variability is therefore a substantive objective choice, not an arithmetic error.

For categorical proposal q_θ with logits $\theta = (\theta_0, \theta_1)$, the direct objective is

$$\mathcal{L}(\theta) = q_\theta(0)c + H(q_\theta) - \log 2.$$

At uniform initialization its gradient is $(c/4, -c/4)$. A single independent return per policy supplies an unbiased score-function estimate of this gradient, although individual updates remain noisy.

To contrast an evidence objective, draw N independent actions and returns, and form

$$W_i = \frac{e^{G_i}}{2q_\theta(A_i)}, \quad \ell_N = \log\left(\frac{1}{N} \sum_i W_i\right), \quad F_N(\theta) = \mathbb{E}[\ell_N].$$

Writing $\bar{W}_i = W_i / \sum_j W_j$ and $s_i = \nabla_\theta \log q_\theta(A_i)$, differentiation gives

$$\nabla_\theta F_N = \mathbb{E}\left[-\sum_i \bar{W}_i s_i + \ell_N \sum_i s_i\right]. \quad (\text{A.9})$$

The first term differentiates the importance weights; the second accounts for sampling actions from the parameterized proposal. This formula is not the direct expected-return gradient. It is also not asserted to equal the gradient of a variational KL to the trajectory target at every finite N .

Each independent draw has three possible outcomes, with probabilities $q_\theta(0)$, $q_\theta(1)/2$, and $q_\theta(1)/2$. Enumerating all 3^N combinations therefore computes Equation (A.9) without Monte Carlo error. At uniform logits, the variable-action component is:

c	N	Evidence gradient	Direct ELBO gradient
0	1	0.000000	0.000000
0	2	0.054223	0.000000
0	4	0.045058	0.000000
0.2	1	-0.050000	-0.050000
0.2	2	0.023860	-0.050000
0.2	4	0.028336	-0.050000

A positive component initially increases preference for the variable action. For $N = 1$, taking the expectation of $G - \log 2 - \log q_\theta(A)$ recovers the direct ELBO on this instance. That special identity does not equate the objectives for larger N , or justify multi-state entropy approximations.

The executable check compares the one-step differentiable implementation with this enumeration for $N = 1, 2, 4$, both values of c , and uniform or nonuniform logits: twelve cases and 96 saved numeric arrays. Independent inspection verifies the outcomes, masses, gradients, source identity and compute record. Native float32 discrepancies remain below the predeclared 3×10^{-6} absolute tolerance. Cross-platform re-summation of derived float64 quantities is checked separately at a 64-machine-epsilon bound; the native tolerance is not enlarged. The largest averaged native-gradient discrepancy is 3.70×10^{-8} .

This check isolates the one-step objective distinction. It does not prove that either update learns a maze better, explain a full-system performance gap, or establish the scalability of either inference engine. Those questions require separately controlled, replicated experiments.

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בחירת מדיניות באמצעות הסקה לפי תוחלת התגמול

Eden Havri

עבודת גמר לתואר מוסמך למדעי הטבע

אוניברסיטת בן-גוריון בנגב

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תקציר

אנו חוקרים הסקה בייסיאנית מוכללת מעל מדיניות דטרמיניסטיות מלאות בתהליך החלטה מרקובי סטוכסטי בעל אופק סופי. התפלגות המטרה נקבעת לפי תוחלת התגמול: תחילה מחשבים תוחלת ביחס לאקראיות בסביבה, ואז מפעילים פונקציה מעריכית. הגדרה זו, המבוססת על עבודות קודמות, שונה ממתן משקל מעריכי לתגמול שהתקבל במסלול אקראי יחיד.

השיטה המעשית הנבחנת לומדת התפלגות מפורקת מעל טבלאות מדיניות מלאות. אלגוריתם PPO מספק אתחול בעל עלות מדווחת בנפרד. לאחריו מתבצע עדכון וריאציוני ישיר בעזרת תגמולים מהסימולטור, הפחתת שונות ואיבר אנטרופיה המחושב בדיוק. אנו מצדיקים את אומד הגרדיאנט ומשווים את תוחלתו לחישוב מדויק בבעיות קטנות. מסלול אחד לכל מדיניות נדגמת מספיק לתקינות האומד, אך חלוקת התקציב בין מדיניות שונות לבין מסלולים חוזרים משפיעה על הלמידה. בכל עדכון נדגמות מדיניות חדשות; בזמן ביצוע נשמרת המדיניות שנבחרה לכל אורך האפיזודה. אין בשיטה זו שלב של שכפול חלקיקים.

התכנון הניסויי מבחין בין תקינות הגרדיאנט, שגיאת משפחת הקירוב וביצועי הבקרה. תקציב האינטראקציות כולל גם את האתחול וגם את העדכון הווריאציוני. ההערכה מפרידה בין התחייבות למדיניות מלאה לבין הגרלת פעולה מחדש בכל ביקור. ניסויי פיתוח ראשוניים בסביבות רשת מצביעים על תועלת לדגימת יותר מדיניותות במקום חזרה מרובה על כל מדיניות. בניסוי הקשה, הקצאה קטנה מדי לשלב החקירה לא יצרה מסלולים מוצלחים. חלוקה מחדש של אותו תקציב שיפרה את הביצוע, אך לא ביטלה את הפער בין כללי הביצוע. בניסוי המשך של שש עשרה הרצות באשכול האוניברסיטה, אחת משלוש החזרות במפה הבינונית נכשלה בלמידה. בקרת PPO מסייעת בחקירה השיגה תגמול גבוה יותר בשבע מתוך שמונה השוואות מזווגות. בניסוי המשך מוגבל, הארכת האתחול באותו תקציב כולל תיקנה את המקרה שנכשל ומנעה קריסה בשתי חזרות חדשות. עם זאת, בשתייהן בקרת PPO השיגה תגמול גבוה יותר בהגרלת פעולות מחדש. בדיקה מצומצמת זו אינה מוכיחה יציבות כללית. אלו תוצאות פיתוח, ולא הוכחה לעליונות כללית.

העבודה כוללת גם שיטות דגימה חלופיות, בדיקות תקינות ותיעוד של ניסיונות שלא הצליחו. דוגמה קטנה עם פתרון מדויק מפרידה בין שגיאת משפחת הקירוב לבין איכות הבקרה. שיפור יציבות האתחול, השוואה מלאה מול שיטות הבסיס ומפות שלא שימשו לפיתוח עדיין נדרשים להשלמת המסקנות.



אוניברסיטת בן-גוריון בנגב

הפקולטה למדעי הטבע

המחלקה למדעי המחשב

בחירת מדיניות באמצעות הסקה לפי תוחלת התגמול

חיבור זה מהווה חלק מהדרישות לקבלת התואר מוסמך למדעי הטבע

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